

# qf-pgpe: a verified pure-Rust projected Gross–Pitaevskii engine for quantum-fluid experiments, with a Python extension

The quantum-fluids module of rusty-SUNDIALS — design, cross-language verification, performance, and what porting found

Xavier Callens

SocrateAI-Scientific-QuantumFluids and rusty-SUNDIALS

<https://github.com/xaviercallens/rusty-SUNDIALS>

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## Abstract

Classical-field (projected Gross–Pitaevskii) simulations of two-dimensional Bose fluids are usually scripts: an FFT library, a time stepper and analysis code whose correctness is checked, when at all, by a few conservation laws. We describe **qf-pgpe**, the quantum-fluids module of the rusty-SUNDIALS numerical library: a pure-Rust integrating-factor RK4 engine for the projected equation on a periodic grid, a vortex instrument (periodic theta-function imprint, sub-grid detection, tracking), the registered estimators of vortex friction and diffusion, a thermal-state toolkit and a vortex–phonon scattering instrument, plus a Python extension that exposes them on numpy arrays. Every module is a port of a Python original of an ongoing, pre-registered research programme and carries a numeric cross-check against it: estimators agree to  $2 \times 10^{-14}$  on real vortex tracks, a 31-run scattering matrix to  $5 \times 10^{-11}$ , measurements on a shared thermal field to  $10^{-14}$ ; the engine is checked against independent known answers and against rusty-SUNDIALS’ own CVODE (global error falling as  $\Delta t^{4.00}$ ). The cross-checks found three defects, one of them in the Python programme: its primary estimator of the mutual friction is biased low by vortex diffusion ( $-8$  to  $-22\%$ ), a flaw the registered synthetic gate had not shown. On a single thread of a 2013 laptop CPU the engine is  $4.3\times$ ,  $4.2\times$ ,  $2.7\times$  and  $2.5\times$  faster per step than the numpy engine at  $N = 64, 128, 256, 512$  ( $3.2\times$ – $1.5\times$  against the fastest of the Python engines we tried, torch on CPU), and a drop-in replacement of the Python stepper makes an end-to-end thermal-state measurement  $5.3\times$  faster with identical results. We make no claim about GPU codes or other published solvers, which we did not measure. *Version 2* adds the first reproduction of an external, published solver run (Kwon and Shin’s flow past a penetrable obstacle, a different equation, scheme and precision): the initial field is reproduced to  $2.8 \times 10^{-8}$ , the force on the obstacle to  $2 \times 10^{-4}$  over 10 time units and the vortex counts at 9 of 10 times to  $t = 50$ , with the places where it does not agree stated; a JAX/XLA row (equal checksums to  $10^{-13}$ ; faster than the single-threaded Rust step at  $N \geq 256$  only because it uses threads), and an intra-step threading option of the engine (bit-identical,  $2.4\times$  at  $N = 512$  on a loaded machine).

## 1 Introduction

Finite-temperature quantum-fluid physics in two dimensions — the Berezinskii–Kosterlitz–Thouless transition, the friction on a quantized vortex, the screening by thermal vortex pairs — is routinely studied with the projected Gross–Pitaevskii equation (PGPE) [1, 2]: a microcanonical classical field  $i\partial_t\psi = P[-\frac{1}{2}\nabla^2\psi + g|\psi|^2\psi]$  with a sharp momentum cutoff  $P$  that makes the thermal cloud, the condensate and the vortices dynamical. The numerical core is small (a 2D FFT pair and a Runge–Kutta stepper); what is large, and error-prone, is what sits on top of it:

imprinting vortices on a torus, finding and following them, reading transport coefficients out of the tracks, measuring temperature and normal density of a state.

This paper describes the module of the rusty-SUNDIALS library [3] (a Rust implementation of the SUNDIALS solver suite) that carries this stack, `qf-pgpe`, and records what we can and cannot say about it. Its design rule is that no module is accepted without a direct numeric comparison against the original it replaces, on the same inputs. Our contributions are: (i) the engine and instruments (Section 2); (ii) a verification record that includes independent integration by the library’s own CVODE and cross-language comparisons with quantified differences (Section 3); (iii) the three defects this found (Section 4); (iv) measured performance against numpy, scipy and torch, with the conditions stated (Section 5); (v) a Python extension with a drop-in stepper (Section 7); (vi) in this version, a reproduction of an external published run (Section 6). The physics results that the programme obtained with these tools are published separately [4, 5, 6]; here they serve as use cases (Section 8).

**What is claimed, and what is not.** Claimed: agreement with the Python originals at the stated levels; agreement with one external published run (Section 6) at the stated levels, with the disagreements listed; single-thread speed-ups over the numpy, scipy and torch-CPU engines on the stated machine; bit-identical results between optimisations and thread counts. **Not claimed:** that this is the best PGPE solver. We did not benchmark GPU codes, FFTW-based C codes or other published solvers (none were installed), no GPU or TPU has been used yet, and the machine is a 2013 quad-core laptop CPU shared with other jobs.

## 2 Design

**Engine.** `ComplexField2D` integrates the projected equation on an  $N \times N$  periodic grid of side  $L$  ( $\hbar = m = 1$ , interaction  $g$ ) by the integrating-factor RK4 scheme: the linear part is exact in Fourier space, the cubic part is classical RK4, the projector is a sharp circle  $|k| \leq k_{\text{cut}} = f k_{\text{max}}$  (default  $f = 1/2$ , the cutoff at which the cubic term is alias-free in the interior of the retained set). FFT conventions are numpy’s (`fftfreq` ordering; unnormalised forward transform), so arrays are exchanged with the Python programme without conversion. Norm, energy and momentum are provided; `rhs/pack/unpack` expose the right-hand side as an ODE system for external integrators.

**Instruments.** Table 1 lists the modules. All are ports of the Python programme’s scripts, with the same formulas and the same registered conventions.

**Performance engineering.** The step is allocation-free (a workspace is allocated once per run, two ping-pong buffers carry the state) and the 2D FFT is rows–transpose–rows with a tiled in-place transpose (unit stride in both passes). In the nonlinear term the FFT lines that are identically zero in a projected field (rows with no mode inside the projector on the way in) or are discarded by the projector (on the way out) are not transformed; this is exact (a test asserts equality bit for bit for three cutoffs) and gave 8–11 %. A rayon-parallel pass over rows was implemented and removed: on this four-core machine it gave no speed-up at  $N = 512$  and 1024 with two and four threads, and independent runs — the unit in which the programme parallelises — scale across processes (Section 5).

## 3 Verification

**Known answers.** The engine carries the programme’s pre-registered known answers: norm conservation ( $10^{-8}$ ), momentum conservation ( $10^{-10}$  for states with weight away from the cutoff

| module     | content                                                                                                                                                                  | Python original              |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| lib        | IF-RK4 PGPE, cutoff fraction, invariants, rhs/pack/unpack                                                                                                                | pgpe.py                      |
| vortex     | periodic theta-function imprint with Bernoulli amplitude, raw plaquette detection with sub-grid refinement, tracker, tracked runs, placements                            | vortex_transport.py          |
| transport  | estimators of $\alpha$ (energy, regression), $1-\alpha'$ , $\eta$ (residual MSD), block jackknife; point-vortex velocity/energy on the torus; synthetic Langevin gate G0 | transport_estimators.py      |
| thermal    | seeded RNG, random_state, heat, thermometer, condensate fraction, current correlators ( $n_s/n$ ), vortex count, Fourier resampling, run_blocks, make_base               | observables.py,<br>round2.py |
| scattering | Bogoliubov-wave dressing of a vortex pair, wave momentum flux, drift slopes, $\sigma_{\parallel}(k)$ , $\sigma_{\perp}(k)$ ; parallel resumable scan runner              | vortex_wave_scattering.py    |
| qf-pgpe-py | Python extension (pyo3, numpy)                                                                                                                                           | —                            |

Table 1: The modules of the quantum-fluids crate. Not ported (declared): the  $g_1$  correlation fits, vortex–dipole matching, the Onsager order parameter, persistent-homology instruments and the symbolic series of the programme.

edge), the exact plane-wave solution ( $10^{-9}$ ), and, for the vortex instrument, the  $T = 0$  gate of the campaign (a neutral pair neither shrinks nor changes momentum; momentum 0.96–0.99 of  $2\pi nd$ ; exactly two detections).

**An independent integrator: CVODE.** The right-hand side of the projected equation, on the retained modes, is an ODE system of dimension  $2N_{\text{modes}}$ . We hand it, unchanged, to rusty-SUNDIALS’ variable-order Adams solver with tight tolerances ( $N = 16$ ,  $L = 16$ , 49 modes,  $t = 1$ ) and use the result as an independent reference. The global error of the IF-RK4 step falls as  $\Delta t^{4.00}$  over  $\Delta t = 0.04 \dots 0.0025$  (Fig. 1, left), which is the order the programme registered (its “K2” check, previously not ported). The test depends on a fix of the library (the Adams order was stuck at one): without it the solver collapses its step on this oscillatory Hamiltonian problem and fails after about  $10^6$  right-hand-side evaluations; with it, 482 evaluations in 95 steps. The test is therefore also a regression guard for the solver.

**Cross-language agreement.** Table 2 gives the measured differences between each Rust module and its Python original on the same inputs.

**A statistical comparison that did not meet its criterion.** For the thermal-state constructor ( $N = 64$ ,  $L = 32$ ,  $e = 0.6$ , nine seeds each) the temperatures are 0.0757 (Rust) and 0.0865 (Python), per-seed standard deviations 0.018 and 0.010: a 12.5 % gap at  $z = -1.6$ , outside the 10 % criterion first set and not significant; a 60-seed comparison at another configuration cannot distinguish the two (vortex count 0.67 against 0.57, condensate fraction  $0.739 \pm 0.009$  against  $0.727 \pm 0.014$ ). The test asserts  $|z| < 2.5$ ; we report the gap rather than the criterion.

**Platform dependence.** The scattering cross-check passes at  $10^{-13}$  on x86-64 Linux, where its numpy fixtures were made, but its  $T = 0$  control run differed by  $8 \times 10^{-3}$  in position on an Apple-silicon CI runner: a pair near the unstable saddle at  $d = L/2$  amplifies rounding differences between libm and FFT kernels. The tolerances are strict on the reference platform and relaxed elsewhere.

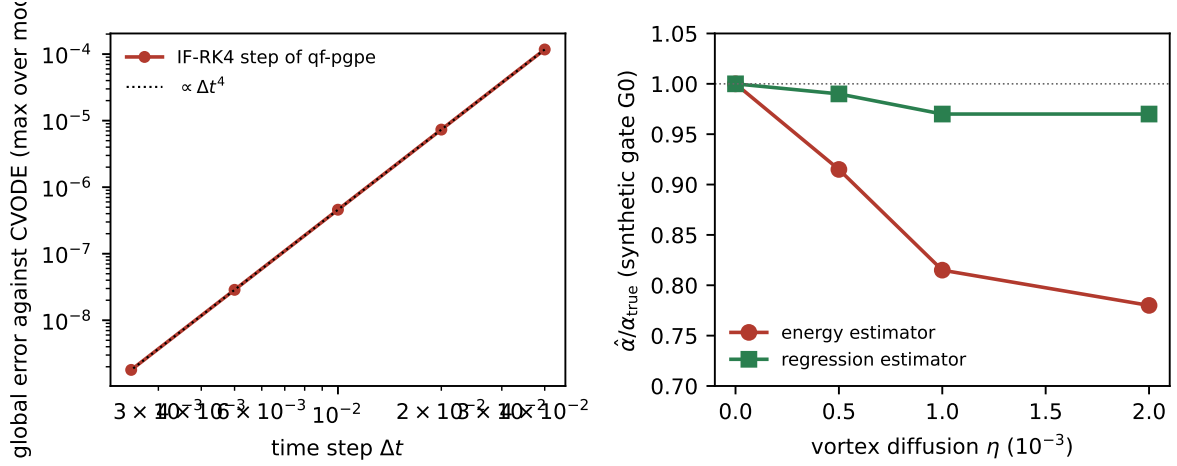


Figure 1: Left: global error of the IF-RK4 step against a CVODE reference of the same right-hand side. Right: the energy estimator of the friction is biased low by diffusion in the registered synthetic gate (a flaw found by the port, Section 4).

## 4 What porting found

1. **A defect in the first Rust version**, found by the vortex cross-check: a vortex centred exactly on a grid node was imprinted with unit modulus instead of zero. Fixed; the separation now agrees to  $10^{-4}$ .
2. **A bias in the Python programme’s friction estimator**. The port of the estimators was bit-identical to Python on real tracks; running the registered synthetic gate over many seeds then showed that the energy estimator of  $\alpha$  is biased low by vortex diffusion: for a true  $\alpha = 0.02$  it returns 0.0200 at  $\eta = 0$  and 0.0183, 0.0163, 0.0156 at  $\eta = 5 \times 10^{-4}$ ,  $10^{-3}$ ,  $2 \times 10^{-3}$  (−8, −18, −22 %), independently of the detection noise, while the regression estimator is unbiased (Fig. 1, right). The registered gate had passed on a single seed (its criterion passes on 2 of 12). The programme’s published friction values were corrected [4, 5].
3. **A hidden dependency of the solver library**: the Adams-order defect above, exposed by the first Hamiltonian system run through CVODE.

## 5 Performance

**Method.** The problem is identical in every engine:  $N \times N$  grid,  $L = N/2$ ,  $g = 1$ ,  $\Delta t = 0.01$ , cutoff  $k_{\text{max}}/2$ , one fixed initial state; the final states of all engines agree to  $1\text{e-}13$  (relative difference of the checksum  $\sum |\text{Re } c| + |\text{Im } c|$ ), so the timings are for the same computation. Engines: numpy (pocketfft) [7], scipy.fft with one and four workers [8], torch on CPU with one and four threads [9], and qf-pgpe on one thread. Statistic: the best of the repetitions (three and five, in two measurement sessions) of the time per step, taken for every engine in the same way. **Machine and conditions:** Intel Core i7-4930MX (2013, 4 cores/8 threads, AVX), Linux, no GPU; during the measurement an unrelated four-thread training job of another project was running, so absolute times are pessimistic and the best-of statistic is used for all engines alike. JAX could not be imported in this environment, FFTW-based C codes and GPU codes were not available: **no comparison with them is made**.

**Results.** On one thread qf-pgpe takes 0.62, 2.73, 22.52 and 104.60 ms per step at  $N = 64$ , 128, 256, 512, against 2.63, 11.50, 60.22 and 260.00 ms for numpy: speed-ups of 4.3, 4.2, 2.7 and 2.5. Against the fastest Python engine at each size (torch on CPU, which is multi-threaded in its

| module     | comparison (same inputs)                                                                                                  | max difference                                                  |
|------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| transport  | four real vortex tracks (800 samples), every estimate and error, non-default options                                      | $2 \times 10^{-14}$ relative (two estimators bit-identical)     |
| thermal    | one shared thermal field ( $N = 32$ ): norm, energy, $T$ (two windows), condensate fraction, $J_L$ , $J_T$ , vortex count | $\leq 3 \times 10^{-14}$ ; count exact                          |
| thermal    | the same field after 20 time units                                                                                        | $2.7 \times 10^{-12}$                                           |
| scattering | short case, three runs: positions, momentum, energy, slopes                                                               | $7 \times 10^{-14}$ , $8 \times 10^{-13}$ , $3 \times 10^{-14}$ |
| scattering | full probe ( $L = 64$ , $N = 128$ , 400 time units): $\sigma_{\parallel}$ , $\sigma_{\perp}$                              | identical to six digits (2.03913, 1.58133)                      |
| scattering | the registered matrix (31 runs, 15 points)                                                                                | $4.6 \times 10^{-11}$ relative                                  |
| vortex     | imprint at positions with a vortex on a grid node; $T = 0$ separation at $t = 5, 40$ ( $N = 64$ )                         | $10^{-4}$ (after a defect fix)                                  |
| extension  | run 20 time units ( $N = 64$ ); <code>imprint_v2</code> ; <code>detect</code> ; <code>analyse_tracks</code>               | $10^{-10}$ ; $10^{-11}$ ; identical; bit-identical              |

Table 2: Rust against Python on identical inputs. The random-number generators differ (xoshiro256\*\* against numpy’s PCG64), so constructions that draw random numbers (`random_state`, `heat`) are compared statistically, measurements on a given field exactly.

four-thread form) the speed-up is 3.2, 2.1, 1.6 and 1.5. The advantage shrinks with  $N$ . We did not profile the origin of the advantage; the step is dominated by FFTs, and the optimisations we measured separately (allocation-free loop, unit-stride transforms, pruned transforms) are each of the order of ten to seventy per cent.

**JAX/XLA and threading (added in version 2).** The same problem, written for JAX [10] (`jit`, `fori_loop`, complex128, CPU, default thread pool), reproduces the Rust checksum to  $\leq 10^{-13}$  at the four sizes, so its times are for the same computation: 1.13, 4.9, 13.6 and 73.6 ms per step at  $N = 64, 128, 256, 512$  against 0.60, 2.9, 27.2 and 123 ms for the single-threaded Rust step measured in the same minutes (machine at load  $\approx 7$ , so every number is perturbed; the same JAX case gave 8.8 ms at  $N = 128$  in an earlier session). XLA is thus faster at  $N \geq 256$ . Pinned to one core (two cores tried) the order reverses: Rust 3.0–3.5, 27.6–30.3, 149–160 ms against JAX 5.4–5.8, 33.6–33.8, 155–172 ms at  $N = 128, 256, 512$ . The XLA advantage is therefore threading, not a better algorithm. The Rust engine now has an optional threading mode (cargo feature `parallel`): row FFT passes, transposes and element-wise loops run on a thread pool; results are bit-identical for any thread count (tested). On the same loaded machine the step at  $N = 512$  takes 160, 120, 88, 68 ms with 1, 2, 4, 8 threads (2.4 $\times$ ; JAX: 74 ms), at  $N = 256$  30.7, 23.8, 19.1, 18.9 ms (1.6 $\times$ ; JAX: 13.6 ms), and at  $N = 128$  threading is slower than serial and must stay off. These are first measurements, to be repeated on an idle machine and on GPU/TPU hardware (JAX is the path to TPUs and the differential-testing oracle for any GPU kernel).

**Throughput and end-to-end use.** The programme’s experiments are many independent runs (seeds, pair sizes, wavenumbers), so throughput is the relevant quantity: with no interpreter lock and no per-step round trip, one process delivers 300 steps per second at  $N = 128$  (numpy: 42) and the best aggregate over the concurrency we tried is 583 against 120, a ratio of 4.9 (Fig. 3). End to end: a 50-time-unit tracked  $T = 0$  vortex run at  $128^2$  takes 41 s of CPU in Rust against 106 s in numpy (2.6 $\times$ ), and a thermal-state measurement (`run_blocks`, 60 time units with measurements,  $N = 64$ ) takes 20.8 s with the drop-in extension against 109.8 s (5.3 $\times$ ) with the same  $T$ ,  $n_s/n$ , vortex count and energy (the measurement code is the same Python).

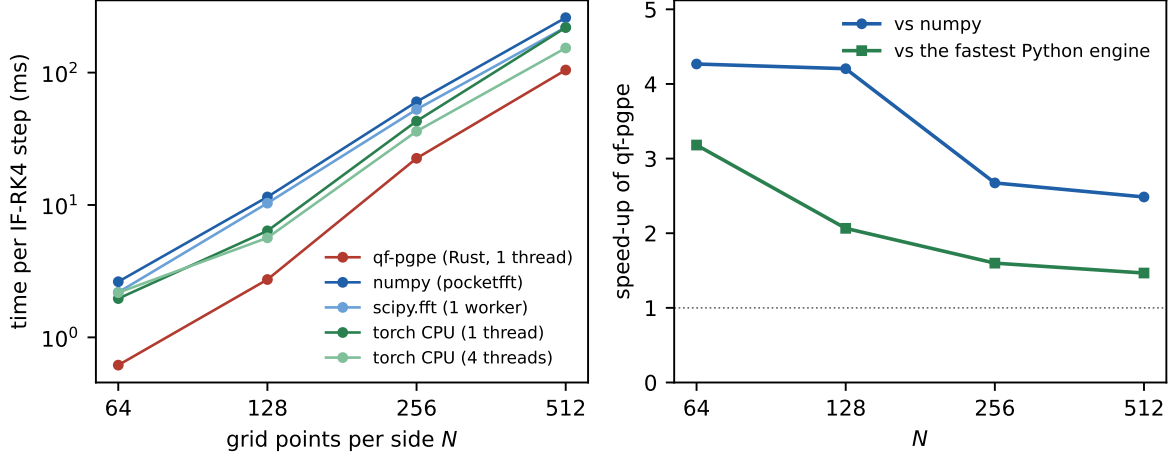


Figure 2: Time per IF-RK4 step against grid size for the engines of the benchmark (left) and the speed-up of qf-pgpe on one thread (right).

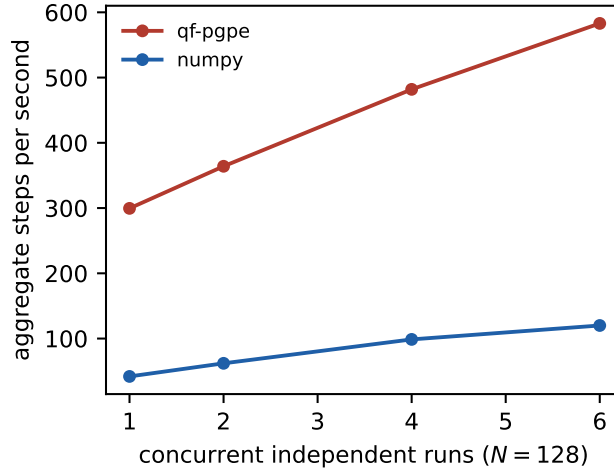


Figure 3: Aggregate throughput of independent runs ( $N = 128$ , one thread each) against the number of concurrent processes, on the same machine and under the same external load.

## 6 Reproducing an external solver run

Everything above compares the engine with its own origin. The first comparison with someone else's published run is the reference run of Kwon and Shin [11] (flow of a superfluid past a penetrable Gaussian obstacle; data and GPU code deposited with a CC-BY licence). It is a different problem from the programme's: a  $1000 \times 500$  grid on  $[-250, 250) \times [-125, 125)$ , an obstacle  $V_0 = 0.9$ ,  $\sigma = 20$ , a moving-frame term, absorbing layers, dissipative dynamics  $\partial_t \psi = v(t) \partial_x \psi - (i + \Gamma) [-\frac{1}{2} \nabla^2 + |\psi|^2 - 1] \psi - iV\psi$ , a split-step scheme in single precision. We implemented that model with an independent scheme (explicit RK4 on the whole right-hand side, spectral derivatives, double precision,  $\Delta t = 0.01$ ) twice, in numpy and in Rust (qf-pgpe::flow); the two agree with each other to the digits quoted. No acceptance threshold was registered for  $t > 10$ ; the results are reported as found.

**What the comparison showed about the reference.** (i) The reference reads the flow velocity at the end of each step; during the 0.1 ramp this shifts the obstacle by 0.00275 and

| quantity                                               | result                                                                                                                                                              |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| initial field                                          | reproduced to $2.8 \times 10^{-8}$ (relative $L^2$ , float32 round-off), including the seeded noise                                                                 |
| force, $t \leq 0.3$ / $\leq 5$ / $\leq 10$             | $\max  \Delta F  = 1.1 \times 10^{-6}$ / $1.4 \times 10^{-4}$ / $2.2 \times 10^{-4}$ ( $ F  \leq 5.5$ )                                                             |
| force, $10 \leq t \leq 50$                             | $7.9 \times 10^{-3}$ then $\approx 1.7 \times 10^{-2}$ (0.19% of the maximum 8.94), saturating after the wake forms                                                 |
| $\psi$ against the snapshots, $t = 10, 20, 30, 40, 50$ | $2.2 \times 10^{-4}$ , $5.8 \times 10^{-4}$ , $1.3 \times 10^{-3}$ , $2.5 \times 10^{-3}$ , $4.1 \times 10^{-3}$ (density: $2\text{--}4 \times 10^{-4}$ throughout) |
| vortex counts, $t = 5, \dots, 50$                      | ours 0, 0, 0, 0, 2, 3, 2, 4, 7, 7; reference 0, 0, 0, 0, 2, 3, 2, 4, 6, 7 (reference rule applied to our snapshots; first shedding at $t = 25$ in both)             |
| own temporal error                                     | $3 \times 10^{-10}$ in the force ( $\Delta t$ halved, $t \leq 5$ )                                                                                                  |

Table 3: Rust `qf-pgpe::flow` against the reference run of Kwon and Shin.

changes the force by  $3.5 \times 10^{-3}$  relative to the stage-time reading — an  $O(\Delta t)$  convention that we reproduce on request (and that makes a halving of  $\Delta t$  an invalid convergence test). (ii) Its stored initial field is the Thomas–Fermi field after *one* imaginary-time step plus noise: with its own tolerance ( $\epsilon = N_x N_y 10^{-9}$ ) the stopping criterion is met at the first step, as both of our implementations confirm. This is a statement about the deposited data, not about the published analysis, whose observables are measured at late times.

**Where it does not agree.** The force separates from the reference once the wake exists, by a factor  $\approx 35$  between the windows 5–10 and 10–20, then saturates; the single count that differs (7 against 6 at  $t = 45$ ) occurs while the count is growing, and we did not examine which vortex is responsible. We did not separate the effect of single precision from that of the discretisation, did not compare beyond  $t = 50$ , and did not compute the Strouhal number. The model was recovered from the code, not from the paper’s text. A CI workflow (`reproduction.yml`, weekly and on demand, outside the required checks because it needs an external download) re-runs the ground state and the first 0.3 time units against a checksummed copy.

## 7 The Python extension

The extension `qf_pgpe` (pyo3 and the numpy crate) exposes `Pgpe` (`run`, `step`, invariants, `imprint_v2`, `detect`, `momentum_band`) and `analyse_tracks` on C-contiguous `complex128` arrays of the numpy layout; `run` evolves for a given time inside one call with the interpreter lock released. A subclass of the programme’s Python stepper replaces only the stepping:

```
from pgpe_rust import PGPERust as PGPE      # every script that did 'from pgpe import PGPE' now steps in
    Rust
s = PGPE(N=128, L=64.0); c = s.run(c, 100.0) # callbacks keep their contract (chunked evolution)
```

so the measurement code is unchanged. A smoke test (the exact plane wave, conservation, imprint and detection, input validation) accompanies the crate.

## 8 Use in the research programme

The tools have been used in a pre-registered programme whose results are published separately: the vortex-transport paper [4] (friction, diffusion and screening of vortices in a closed field), the friction-law paper [5] (the friction follows the temperature, not the normal density), and the software record [6]. Two uses are specific to this module. First, the 31-run vortex–phonon scattering matrix was reproduced independently in Rust, point by point, to  $5 \times 10^{-11}$ ; the follow-up with a larger box (amendment WS-A2) and the replication of the pair-separation

matrix are run on it. Second, the estimator bias of Section 4 would not have been found by re-running the same code: it needed a second implementation and a seed sweep that was cheap enough to run.

## 9 Limitations and future work

- One machine, one CPU generation, a concurrent external load; no GPU or TPU yet, one external run reproduced, no comparison with other published solvers or with FFTW-based C codes. The plan (a reference-solver gap analysis, a reproduction registry in CI, a differential-testing design with JAX as the TPU path) is in the programme’s repository; none of its GPU/TPU items is measured.
- Threading inside a run helps at  $N \geq 256$  but the scaling is only measured on a loaded machine and is below that of XLA at  $N = 256$ ; below  $N = 256$  it must stay off.
- The thermal constructors use their own RNG; reproducing a numpy-seeded construction bit for bit is not possible and not a goal.
- The module is two-dimensional and periodic; the programme’s three-dimensional and trapped geometries are not covered.
- Momentum is conserved to  $\sim 10^{-6}$  (not  $10^{-10}$ ) for noisy fields with weight at the cutoff edge, where the cubic term aliases; the numpy engine shows the identical drift.

## 10 Availability

Source: <https://github.com/xaviercallens/rusty-SUNDIALS> (crates `qf-pgpe`, `qf-pgpe-py`; merged as pull requests #69 and #70; the reproduction and threading of version 2 are in the follow-up branch `feat/qf-pgpe-reproduction`); BSD-3-Clause. The Python programme, the benchmark script (`exploration/pgpe/bench_engines.py`), the data (`data/generated/pgpe/bench/`) and the ledger are in [6]. Reproduction: `cargo test -release -p qf-pgpe` (including the threaded bit-identity test), `python3 exploration/pgpe/bench_engines.py single`.

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